recovery_eval: State-Matched and Provenance-Aware Evaluation of Recovery Behavior in Language-Model Reasoning

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Abstract—Evaluating whether post-training enhances a language model's ability to recover from intermediate reasoning errors remains challenging due to confounding variables in trajectory evaluation. In this paper, we introduce recovery_eval, a data-centric evaluation framework that pairs verifier-defined error states with prospectively matched reference control states using frozen structural covariates and append-only exposure governance. We validate the framework across 400 genuine neural rollouts generated by two released checkpoints (Qwen2.5-Math-1.5B Base and Instruct) on 20 synthetic arithmetic problems inspired by GSM8K reasoning under a corrected state-matched protocol (Protocol 2.0; 256-token budget). Under state-matched prefixes, neither model exhibits spontaneous error recovery, achieving exactly 0.00% accuracy across all 200 recovery continuations ($p_{\text{BR}} = p_{\text{IR}} = 0.0000$). Instead, models exhibit continuation inertia: 98.5% of recovery rollouts faithfully propagate the injected error downstream. Exact Clopper-Pearson 95% upper confidence bounds demonstrate that while rollout-level observations bound the true recovery rate below 1.49% ($n = 200$), accounting for problem-level clustering yields an effective upper bound of 13.91% ($C = 20$). On unperturbed control prefixes, both models achieve near-perfect accuracy (Base: 97.00% heuristic, 100.00% structural; Instruct: 100.00%), yielding a matched recovery contrast of $D = 0.0000$ ($SE = 0.0000$) under structural verification and $D = -0.0300$ ($SE_{\text{cluster}} = 0.0164, p = 0.0828$) under heuristic parsing. These findings demonstrate that historical reports of a significant negative post-training interaction ($D \approx -0.21$) were artifacts of prompt length asymmetry, 64-token truncation censoring, and heuristic regex verifiers, establishing the necessity of state-matched evaluation infrastructure for reasoning diagnostics.

Index Terms—Language Models, Mathematical Reasoning, Error Recovery, State-Matched Evaluation, Data-Centric AI, Benchmark Governance, Continuation Inertia.

I. Introduction

Large language models (LLMs) trained on mathematical reasoning tasks exhibit substantial improvements in step-by-step problem-solving [1], [2]. A foundational question in machine learning evaluation is whether post-training procedures (such as instruction tuning, process supervision, or reinforcement learning) instill structural intelligence that enables models to detect and recover from early intermediate arithmetic or logical errors [3], [4].

However, evaluating error recovery behavior is acutely vulnerable to evaluation artifacts. Naive comparisons between error-containing trajectories and unperturbed baselines routinely conflate trajectory depth, remaining solution length, token complexity, and prompt phrasing. Furthermore, tight generation limits induce differential truncation censoring, masking true baseline capabilities. In preliminary experiments using an unbalanced 64-token protocol (Protocol 1.0), an apparent negative interaction was observed ($D \approx -0.21$), suggesting that instruction tuning degraded recovery capability.

To resolve these ambiguities, we present recovery_eval, a reproducible data-centric framework that enforces prospective caliper matching on intermediate reasoning states and maintains end-to-end cryptographic provenance. Across 400 genuine neural rollouts evaluated under a length-balanced, state-matched protocol with an expanded 256-token budget (Protocol 2.0), we establish that:

1) Dissolution of the Negative Interaction: Under state-matched prompting, the historical negative interaction dissolves ($D = -0.0300, p = 0.0828$ under heuristic parsing; $D = 0.0000$ under structural parsing).
2) Continuation Inertia: Small (1.5B) language models do not spontaneously self-correct from injected intermediate errors (0/200 recoveries). In 98.5% of cases, they treat corrupted premises as authoritative and faithfully propagate errors downstream.
3) Finite-Sample Uncertainty Bounds: We establish that observing zero recoveries bounds the true recovery rate below 1.49% at the rollout level ($n = 200$), but cluster-robust modeling reveals an effective upper bound of 13.91% ($C = 20$).
4) Evaluator Brittleness: Heuristic last-token regex extractors artificially penalize base models by 3% due to trailing conversational explanations, whereas structural parsing demonstrates 100% baseline accuracy for both models.

II. Related Work

Self-correction and backtracking in LLMs have attracted intense recent scrutiny [5]–[7]. Table I compares recovery_eval with existing paradigms.

TABLE I
Comparison with Existing Reasoning Evaluation Frameworks

Framework	State Matching	Exposure Ledger	Primitive Provenance
End-to-End GSM8K [1]	None	No	Final Answer Only
PRM800K [3]	Unmatched	Partial	Step Scores
STaR [4]	Unmatched	No	Filtered Traces
recovery_eval (Ours)	State-Matched	Append-Only LEDGER	Full BPE Token + Raw JSONL

Unlike end-to-end benchmarks, recovery_eval prospectively matches recovery states with control states prior to generation, eliminating prefix length and depth confounds [8]–[10].

III. Problem Definition & Estimand

Let s denote a partial reasoning trajectory prefix for an arithmetic problem instance $p \in \mathcal{P}$. A deterministic verifier $\mathcal{V}(s)$ audits intermediate step correctness. A recovery state s_R contains a verifier-identified arithmetic error in Step 1. A reference control state s_C represents an unperturbed valid Step 1 prefix for the same problem at an equivalent problem-solving horizon.

For a target policy π (e.g., Instruct) and baseline policy π_0 (e.g., Base), let $V(s) \in \{0, 1\}$ denote the binary correctness indicator of completing the solution correctly from prefix s. We define the matched recovery contrast D as:

$$D = \mathbb{E}[V_\pi(s_R) - V_{\pi_0}(s_R)] - \mathbb{E}[V_\pi(s_C) - V_{\pi_0}(s_C)] \quad (1)$$

IV. The recovery_eval Protocol

A. State Construction and Injected Perturbation

Recovery states s_R are constructed by introducing controlled single-step arithmetic perturbations into Step 1 prefixes. Reference control states s_C maintain unperturbed Step 1 calculations.

B. Prospective Caliper Matching

To guarantee statistical balance, each recovery state s_R is matched to a control state s_C over structural covariates (trajectory depth, remaining length, and token length) ensuring exact categorical matching on reasoning operation (chained integer multiplication). In Protocol 2.0, Control and Recovery prefixes have identical character and token lengths ($\Delta_{\text{chars}} = 0, \Delta_{\text{tokens}} = 0$).

C. Exposure & Provenance Governance

All evaluation items are governed by an append-only event ledger with SHA-256 parent hash chaining to eliminate data leakage. Raw rollout records store input token IDs, generated BPE token IDs, wall-clock generation durations, and exact model weight manifest digests.

V. Experimental Configuration

The empirical campaign evaluates 20 synthetic multi-step arithmetic problems inspired by GSM8K word problems ($N = 20$). For each problem, 1 matched recovery state and 1 matched control state are evaluated across 2 model checkpoints using 5 stochastic generation seeds ($S = 5$), yielding exactly 400 rollouts ($20 \times 2 \times 2 \times 5 = 400$).

TABLE II
Evaluation Design and Provenance Statistics

Attribute	Specification / Value
Base Model	Qwen/Qwen2.5-Math-1.5B (Rev 4a83ca6e)
Instruct Model	Qwen/Qwen2.5-Math-1.5B-Instruct (Rev aafeb0fc)
Hardware Platform	NVIDIA GeForce RTX 4090 (24GB VRAM, CUDA 12.4)
Execution Precision	Native float16 PyTorch 2.4.0
Generation Config	$T = 0.7$, Top-$p = 0.9$, max_new_tokens = 256
Total Rollouts	400 (200 Base, 200 Instruct)
Total Generated Tokens	11,917 tokens (Mean: 29.79 tok/rollout)
Inference Duration	265.40 seconds (45.05 tokens/sec)
Truncation Rate	0.00% (0 / 400 rollouts truncated)
EOS Termination Rate	100.00% (400 / 400 rollouts hit EOS)
Raw Dataset SHA-256	b8792f8df375fccdbd800d440cc884ef...

VI. Empirical Results

A. Cluster-Robust Factorial Outcomes

Table III summarizes completion accuracies across all four experimental cells under both heuristic and structural verification regimes.

TABLE III
Protocol 2.0 Factorial Outcomes ($N = 400$ Rollouts, $C = 20$ Clusters)

State Condition	Base (π_0)	Instruct (π)	Difference (Δ)
Recovery States (s_R)	0.0000	0.0000	+0.0000
Control States (Heuristic)	0.9700	1.0000	+0.0300
Control States (Structural)	1.0000	1.0000	+0.0000
Matched Contrast (Heuristic)	–	–	−0.0300
Cluster SE / p-value	–	–	$SE = 0.0164, p = 0.0828$
95% Confidence Interval	–	–	$[-0.0643, +0.0043]$
Matched Contrast (Structural)	–	–	+0.0000
Cluster SE / p-value	–	–	$SE = 0.0000, p = 1.0000$

B. Dissolution of Negative Interaction

Under Protocol 2.0, the historical negative interaction dissolves. Under the heuristic parser, $D = -0.0300$ ($SE_{\text{cluster}} = 0.0164, t = -1.831, p = 0.0828$). Because the 95% confidence interval ($[-0.0643, +0.0043]$) crosses zero, the effect is statistically indistinguishable from zero at $\alpha = 0.05$. Under structural arithmetic parsing, $D = 0.0000$ exactly across all 20 problem clusters. The empirical zero variance ($SE = 0.0000$) is a descriptive consequence of ceiling (100%) and floor (0%) bounds in this 20-problem benchmark, rather than an assertion of infinite population certainty.

C. Independent Verifier Falsification

The 3% deficit in Base Control (97% vs. 100%) was audited trace-by-trace. In all three disputed rollouts (problems fresh_004, fresh_006, fresh_013 under seed 405), the Base model calculated the correct revenue $(180, 176, 360)$ but appended a post-solution explanation concluding in "- In step 3...". The heuristic extractor's last-token fallback rule parsed the step index 3 instead of the revenue answer. Under structural equivalence, Base Control accuracy is 100/100 (1.0000).

D. Behavioral Trace Analysis: Continuation Inertia

Across all 200 recovery rollouts:

- Direct Error Propagation (98.5%, 197/200): Models adopted the corrupted Step 1 quantity E and computed $4 \times E$.
- Attention-Slip Divergence (1.5%, 3/200): In gsm8k_fresh_004, the model bound the unit price token into the hourly rate.
- Zero Backtracking (0.0%, 0/200): Zero rollouts contained error-detection words (wait, actually, mistake).

E. Uncertainty Bounds for Zero Recovery

While the observed recovery rate is 0/200 (0.0%), naive binomial models (95% upper bound 1.49%) ignore intra-cluster correlation. Accounting for $C = 20$ problem clusters ($N_{\text{eff}} \approx 20$), the exact Clopper-Pearson 95% upper bound is **13.91%** [11], as detailed in Table IV.

TABLE IV
Clopper-Pearson Exact 95% Confidence Upper Bounds

Aggregation Level	Effective Sample (N)	Observed k	Exact 95% Upper Bound
Rollout Level (Total)	200	0	1.49%
Model Level (Base / Inst)	100	0	2.95%
Cluster Level (Problems)	20	0	13.91%

F. Human Validation and Annotation Gate

To validate automated verification against human judgment, we constructed a stratified blinded packet of 40 rollouts across models and conditions, audited against a six-dimensional annotation rubric. Author adjudication demonstrated 100% concordance with our semantic equivalence verifier, confirming 0% recovery, 100% error propagation, and validating that base-model trailing explanations represent mathematically correct reasoning. To maintain scientific integrity, formal multi-rater crowdsourced inter-rater reliability is registered as an open reproducibility protocol.

VII. Limitations

- Model Scale: Evaluated on 1.5B parameters; behavior of 7B–72B or reasoning-specialized models (e.g., DeepSeek-R1) remains an open empirical question.
- Task Scope: Evaluated on 2-step synthetic arithmetic continuation; does not cover multi-branch logic or interactive theorem proving.
- Decoding Mode: Evaluated under unprompted nucleus sampling ($T = 0.7, p = 0.9$) without external verifier tree search.
- Causal Boundaries: The assigned prefix condition is causally identified, but checkpoint comparisons remain observational.

VIII. Reproducibility & Artifact Availability

All code, raw evidence, and verification scripts are archived. The complete neural dataset 10_COMPLETE_QWEN_PROTOCOL2_ROLLOUTS.jsonl (SHA-256: b8792f8df375fccdbd800d440cc884ef2db2f860ff3533af01c70161a5cc1d15) and evaluation package are preserved in the repository.

IX. Conclusion

We presented recovery_eval, a state-matched evaluation framework for language-model reasoning trajectories. By eliminating prompt length asymmetry and truncation censoring, we demonstrated that prior claims of a negative post-training recovery interaction were artifacts of evaluation design, and established that 1.5B language models exhibit continuation inertia rather than intrinsic self-correction.

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